Comparing MicroPython, Zephyr, Lua, and Forth

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Introduction

It's a powerful thing to be able to interact with your embedded system over a serial port. Even in its most basic form, a simple command-line interface (CLI) that responds to a defined set of commands and parameters can make it easy to quickly prototype on your device, modify configuration values, or inspect its running state, all without needing a debugger.

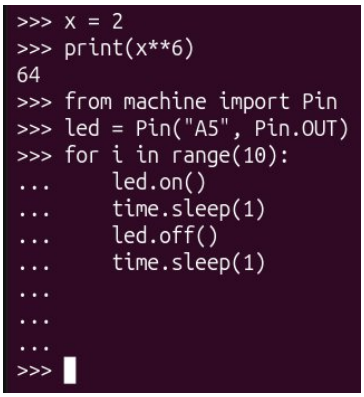
```
uart:~$ help
Please press the <Tab> button to see all available commands.
You can also use the <Tab> button to prompt or auto-complete all commands or its subcommands.
You can try to call commands with <-h> or <--help> parameter for more information.

Shell supports following meta-keys:
  Ctrl + (a key from: abcdefklnptuw)
  Alt  + (a key from: bf)
Please refer to shell documentation for more details.

Available commands:
  clear    : Clear screen.
  date     : Date commands
  device   : Device commands
  devmem   : Read/write physical memory
             Usage:
             Read memory at address with optional width:
             devmem <address> [<width>]
             Write memory at address with mandatory width and value:
             devmem <address> <width> <value>
  hello    : Print a greeting
  help     : Prints the help message.
  history  : Command history.
  kernel   : Kernel commands
  rem      : Ignore lines beginning with 'rem '
  resize   : Console gets terminal screen size or assumes default in case the
             readout fails. It must be executed after each terminal width change
             to ensure correct text display.
  retval   : Print return value of most recent command
  shell    : Useful, not Unix-like shell commands.
uart:~$ hello
Hello from Zephyr shell!
uart:~$ devmem dump -a 0x08000000 -s 16 -w 32
08000000: 00 1f 00 20 05 36 00 08  57 7d 00 08 f1 35 00 08 |... .6.. W}...5..|
uart:~$
```

In fact, one of the many utilities that are provided "out-of-the-box" by the **Zephyr** project is a fully-featured CLI (called the `shell` utility).

Going a step further, though, we could supercharge our CLI if we used an *interpreted language* instead of just a set list of commands. With an interpreted language we could declare new variables and compose loops, conditionals, and functions, all from the command line.



One of **MicroPython**'s best features is its REPL ("read-evaluate-print-loop"), which allows a person to write Python code and interact with their hardware, all from a serial connection. **Lua** and **Forth** are two other interpreted languages that came up in my research as being particularly well-suited for embedded systems.

So, how do these tools stack up against each other? Which one is best for quickly prototyping a new design? We'll take a look at each one, focusing on:

- How easy it is to get to a prompt/REPL (for fully-supported boards and for brand new hardware)
- The compiled size of the final executable
- The difficulty of interfacing with C/C++ code and building a complete application
- Other considerations like the maturity of the library ecosystem, each language's relative speed, their use of the heap, and notable language features

I'll be using a Nucleo-F411 development board to test out each of these projects. Other development boards that can run each of the projects below include: STM32 Nucleo-F401/L476/H743, Nordic nRF52840/nRF52 (including Arduino Nano 33 BLE), NXP FRDM-K64F, Raspberry Pi Pico, and ESP32 DevKitC.

MicroPython

MicroPython (and its close cousin, CircuitPython) are very neat projects that let a developer run Python code on an MCU and, perhaps just as importantly, provide a REPL and a file system out-of-the-box, for extremely quick prototyping.

Getting MicroPython to run on my Nucleo-F411 required:

1. Downloading the MicroPython library and any submodules:
   ```
   # MicroPython source + submodules (CMSIS, ST HAL, etc.)
   git clone https://github.com/micropython/micropython.git
   ```

```
cd micropython
git submodule update --init --depth 1

# Build the host-side cross-compiler (required before any port
build)
make -C mpy-cross
```

2. Optionally copying and changing any settings in `mpconfigboard.h` and `mpconfigboard.mk` for my specific system. (Among other things, I chose to turn off a few libraries that I wasn't going to need, to use single- instead of double-precision floating-point numbers, and to compile at –Os.)
3. For out-of-source builds, I also needed to copy `pins.csv` and `stm32f4xx_hal_conf.h` into my project folder.
4. Use `make` to build the firmware, providing values to `BOARD`, `BOARD_DIR`, and `BUILD` for build files to land in the project folder (not the MicroPython source tree).

```
MPY_DIR  ?= $(HOME)/Documents/CodeLibraries/micropython
BOARD    := F411_MINIMAL
BUILD    := $(CURDIR)/build

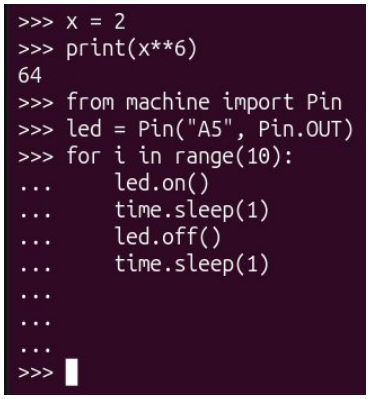
all:
     $(MAKE) -C $(MPY_DIR)/ports/stm32 \
          BOARD=$(BOARD) \
          BOARD_DIR=$(CURDIR)/boards/$(BOARD) \
          BUILD=$(BUILD) \
          MPY_DIR=$(MPY_DIR) \
          -j$(shell nproc)
```

5. Flash the binaries. Ex:

```
openocd -f board/st_nucleo_f4.cfg \
        -c "init; \
            reset halt; \
            stm32f4x mass_erase 0; \
            flash write_image erase $(BUILD)/firmware0.bin 0x08000000; \
            flash write_image erase $(BUILD)/firmware1.bin 0x08020000; \
            verify_image $(BUILD)/firmware0.bin 0x08000000; \
            verify_image $(BUILD)/firmware1.bin 0x08020000; \
            reset run; exit"
```

All told, it only took me a few hours to get going, which wasn't nearly as bad as I'd originally expected! Connecting over serial brought me to a satisfying REPL, where I had the full power of the MicroPython language (a subset of Python) and could import a large

number of libraries that could make developing and doing high-level things exceedingly easy.



The final compiled size (even after my few optimizations) was **277 kB**.

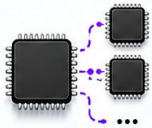
Additionally, my research shows that MicroPython runs about **50-250x slower** than native C/C++, possibly even slower than that for compute-heavy programs, such as when computing FFTs (fast Fourier transforms), SHAs (secure hash algorithm), or FIR (finite impulse response) filters.

——[! NONE of these sizes include floating-point math]————————

The sizes of each compiled binary in this article do not include any software floating-math libraries because the STM32F411 has a *hardware* floating-point unit (FPU). Since essentially every project listed here requires some kind of floating-point math support, however, any device that doesn't have a hardware FPU will incur the extra flash cost of any required software floating-point library functions. Keep that in mind if you build any of these projects for an MCU without a hardware FPU!

However, this was a bit of an idealized scenario in which I didn't have to write *any* new code for my device to be able to run MicroPython; my development board was fully-supported by the library. As you can imagine, I'd have some more work to do if I wanted to run MicroPython on an STM32F411 that was on a custom board, or on an STM32 device from an MCU family that had no port yet (such as the STM32H5). In fact, this work could be *very* non-trivial, taking anywhere from days to ***months*** depending on how much code I have to write or provide from scratch. The graphic below summarizes, for five progressive

scenarios, roughly how many lines of code (LOC) you'd need to write or modify (and in which files) and how long that might take to port MicroPython to the listed device.

Device level	What needs changing	Approx. LOC	Approx. time
Supported board STM32F411 		10–30	Hours–1/2d
Supported MCU STM32F411 		75–150	1/2–1d
Supported MCU family STM32F446, STM32F427 		200–500	1–2d
Supported vendor STM32H5 		500–1500	1–2w
New vendor Nuvoton 		2000–6000	1–8w

Interfacing with C/C++ code (for performance reasons or if you were porting MicroPython to a new device and were writing the HAL to let it interact with the MCU) is straightforward in the simple case: every value in or out is of type `mp_obj_t`, which is a tagged union of possible Python data types, and functions are registered using macros that specify the exact number of arguments the function accepts.

```
// Exactly 1 positional arg
mp_obj_t my_func(mp_obj_t a);
MP_DEFINE_CONST_FUN_OBJ_1(my_func_obj, my_func);
```

```
// Keyword arguments
// args[0..n_args-1] are positional, args[n_args..n_args+2*n_kw-1] are
// key/value pairs
mp_obj_t my_func(size_t n_args, size_t n_kw, const mp_obj_t *args);
MP_DEFINE_CONST_FUN_OBJ_KW(my_func_obj, 0, my_func);
```

You can access the values inside these arguments using conversion functions like `mp_obj_get_int(arg)` or `mp_obj_get_float(arg)`.

If you're writing the HAL, your code has to execute synchronously and you have to match the API that MicroPython expects for each peripheral, which may not be quite how you want to use them. Things also get weird if you want to implement streaming data, need try/catch semantics in your C/C++ code, or are trying to incorporate interrupt-driven code.

One last thing to consider is that MicroPython requires dynamic memory allocation to run.

Zephyr

Zephyr is a "hardware agnostic firmware environment" with the main goal being that a person can write an application once and then re-target it to another piece of hardware (new MCU, new sensors, etc) by changing as little as 1 line of code. Unlike all of the other options I chose to compare, Zephyr does *not* implement anything resembling an interpretable language; however, it does come with a very handy `shell` utility (enabled by changing a single line in a configuration file) that can let you interact with your device over a serial port.

Getting Zephyr to run on my Nucleo-F411 required:

1. Installing West, and then the Zephyr SDK from here (I grabbed the latest minimal release for Linux)
   ```
   pip install west
   mkdir ~/zephyrproject && cd ~/zephyrproject
   west init
   west update

   wget https://github.com/zephyrproject-rtos/sdk-
   ng/releases/download/vX.X.X/zephyr-sdk-X.X.X_linux-
   x86_64_minimal.tar.xz
   tar xf zephyr-sdk-X.X.X_linux-x86_64_minimal.tar.xz
   cd zephyr-sdk-X.X.X
   ./setup.sh
   cmake -P cmake/zephyr_sdk_export.cmake
   ```
   a. I also had some trouble installing dependencies like jsonschema, pyelftools, and libusb, which I eventually fixed by manually installing them:
      ```
      sudo apt install libusb-1.0-0-dev libudev-dev
      ~/.local/share/pipx/venvs/west/bin/python -m pip install \
          -r
      ~/Documents/CodeLibraries/zephyrproject/zephyr/scripts/requi
      rements-base.txt \
      ```

```
    -r
~/Documents/CodeLibraries/zephyrproject/zephyr/scripts/requirements-build-test.txt
```

2. Make `prj.conf` (project-level configuration), `main.c`, and `CMakeLists.txt` (for compiling `main.c` and any other source files).
3. Build and flash

```
export ZEPHYR_BASE=~/zephyrproject/zephyr
west build -b nucleo_f411re -d build .
west flash --runner openocd
```

All told, it only took me a few hours to get going, which wasn't nearly as bad as I'd originally expected! Connecting over serial brought me to a prompt, with some minimal functionality (though the minimalism had more to do with the fact that there was nothing in my application than with the limitations of the Zephyr `shell`).

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uart:~$
```

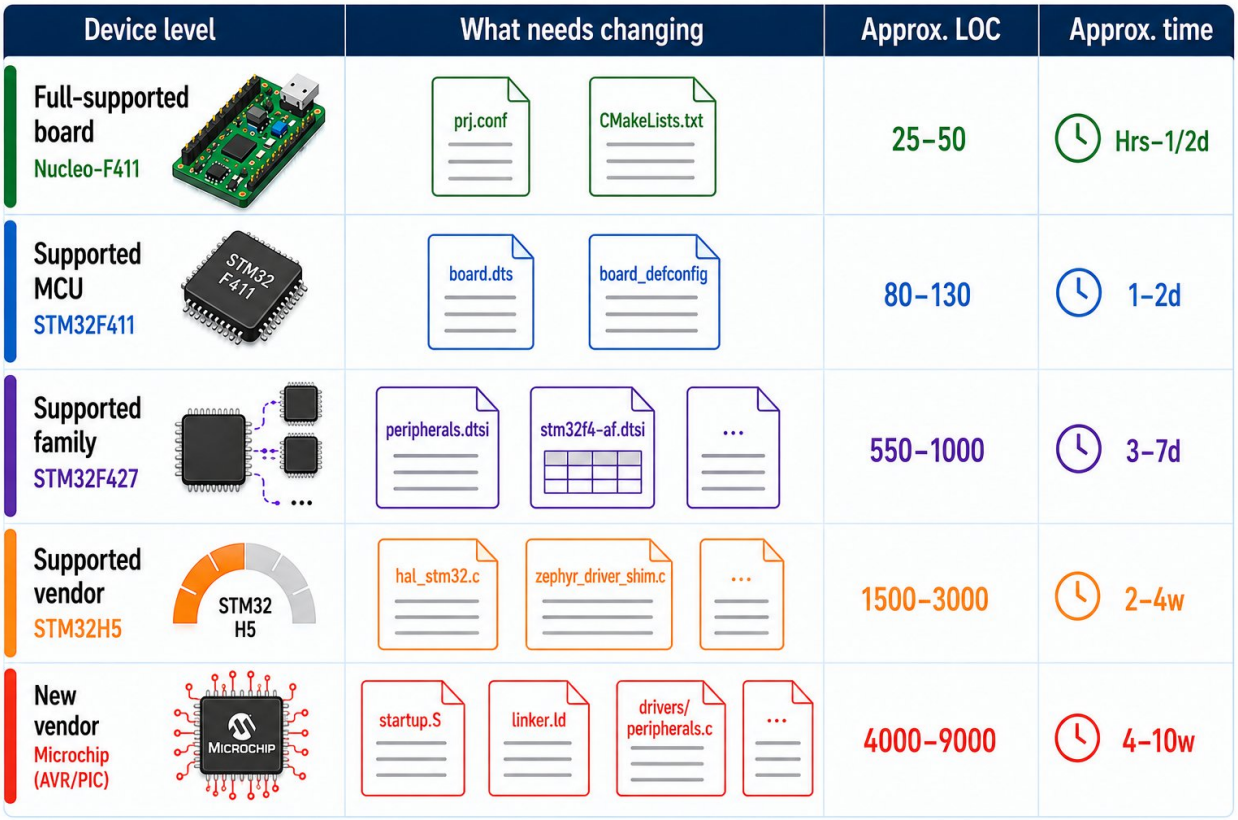
The final compiled size (even after a few optimizations) was **44 kB**.

```
> ls -l build/zephyr/zephyr.bin
-rwxrwxr-x 1 nathancharlesjones nathancharlesjones 45456 May 30 13:14 build/zephyr/zephyr.bin
```

About 30 kB of this is the Zephyr kernel, HAL, and main. The remaining 14 kB can be attributed to the actual `shell` utility.

Additionally, Zephyr is going to run **the fastest** out of all four of these options, since any user code would be written in native C/C++ and executed directly as machine code.

However, as was the case with MicroPython, this was a bit of an idealized scenario in which I basically didn't have to write *any* new code for my device to be able to run Zephyr. Porting Zephyr to a different device, however, could take anywhere from days to ***months***.



In general, porting Zephyr takes slightly more code than an equivalent effort for MicroPython (the DTSI files can be quite verbose, and there are stricter requirements when writing the peripheral drivers for a new vendor) and can be slightly more confusing (how an application maps to the hardware is essentially controlled at build-time by the DTS and DTSI files, not by C/C++ source code, which are configuration files with foreign syntax to anybody who hasn't done Linux kernel development). On the other hand, it's much easier to integrate Zephyr with other C/C++ code; Zephyr *is* in C!

Zephyr peripheral drivers give a developer more control than MicroPython does, by supporting various forms of asynchrony and by exposing access to the peripheral registers when low-level control is needed (as for using a peripheral in a non-standard way, such as UART in LIN mode, or SPI in 3-wire half-duplex).

Lua

Lua is a very interesting language that shares some of Python's high-level feel, which makes writing code in Lua feel rather effortless. For example, here's some Lua code that prints the numbers 1 through 10, that sorts and prints a table, and that implements the start of an "Animal" class:

```
for i=1,10 do print(i) end
```

```
t = {3,1,4,1,5,9}; table.sort(t); print(table.concat(t, ", "));
```

```
local Animal = {}
Animal.__index = Animal
function Animal.new(name, sound)
    local self = setmetatable({}, Animal)
    self.name = name
    self.sound = sound
    return self
end
```

Furthermore, Lua was designed to be embedded in other languages, so it's quite easy to build and to use on embedded systems.

Getting Lua to run on my Nucleo-F411 required:

1. Downloading the Lua source code
   ```
   cd #install location
   curl -L -R -O https://www.lua.org/ftp/lua-5.5.0.tar.gz
   tar zxf lua-5.5.0.tar.gz
   rm -rf lua-5.5.0.tar.gz
   ```
2. Creating an STM32 project in CubeMX
3. Copying `luaconf.h` to `Core/Inc` and slightly modifying it:
   ```
   #define LUA_32BITS
   #undef  LUA_PATH_DEFAULT
   #define LUA_PATH_DEFAULT  ""
   #undef  LUA_CPATH_DEFAULT
   #define LUA_CPATH_DEFAULT ""
   ```
4. Adding the Lua source files (e.g. `lmathlib.c`, `lparser.c`, `ltm.c`, etc) to the project `makefile` (and making changes to a few other lines)
5. Adding a REPL to `main.c`, which amounts to getting a line of text from the user (I'm using the EmbeddedCLI library from Andre Renaud to do that, which also gives us

some basic line editing) and then giving that line to `luaL_dostring` inside the `while(1)` loop.

```
lua_State *L = luaL_newstate();
luaL_requiref(L, "_G",     luaopen_base,   1); lua_pop(L, 1);
luaL_requiref(L, "math",   luaopen_math,   1); lua_pop(L, 1);
luaL_requiref(L, "string", luaopen_string, 1); lua_pop(L, 1);
luaL_requiref(L, "table",  luaopen_table,  1); lua_pop(L, 1);
struct embedded_cli cli;
embedded_cli_init(&cli, "> ", put_char, NULL);
embedded_cli_prompt(&cli);
while(1) {
  if(uart_avail() && embedded_cli_insert_char(&cli,
uart_readc())){
    const char *line = embedded_cli_get_line(&cli);
    if (line && strlen(line) > 0){
      if (LUA_OK != luaL_dostring(L, line)){ /* error */ }
      lua_settop(L, 0);
      embedded_cli_prompt(&cli);
    }
  }
  // Other while(1) stuff
}
```

6. Building and flashing

```
make
openocd -f board/st_nucleo_f4.cfg -c "program $(BUILD_DIR)/\
    lua_in_lua_on_STM32.bin verify reset exit 0x08000000"
```

For detailed setup instructions, see [my Github repository].

The process of building Lua was significantly easier than for MicroPython or Zephyr; there were only a few dozen source files to contend with and there is no required build system to use, so I got to pick my own. Getting up and going took only a few hours; about as long as for MicroPython or Zephyr. After flashing and connecting over serial, I was greeted with a Lua REPL.

```
> x = 2
> print(x^6)
64.0
> for i=1,10 do print(i) end
1
2
3
4
5
6
7
8
9
10
> t = {3,1,4,15,9}; table.sort(t); print(table.concat(t, ", "))
1, 3, 4, 9, 15
>
```

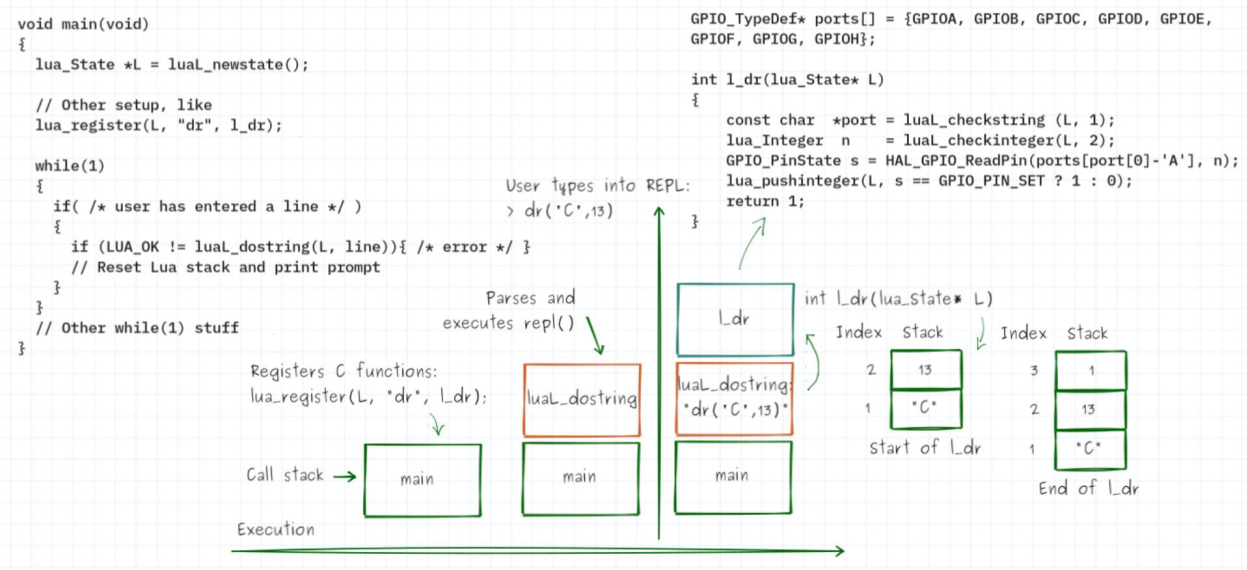
The final compiled size (with a few minor size optimizations) was **144 kB**, of which about 128 kB is from Lua (the rest being the STM32 HAL, C library functions, and main).

```
> ls -l build/lua_on_STM32.bin
-rwxrwxr-x 1 nathancharlesjones nathancharlesjones 147340 Jun 11 10:30 build/lua_on_STM32.bin
```

In my research, Lua seems to run about **10-50x slower** than native C/C++ code, which is also about **5x faster than MicroPython**.

There's no built-in HAL, like there is with MicroPython, but this is easily remedied by writing our own and adding it to the Lua environment. We can do this by:

1. Writing a C/C++ function to interact with our hardware, such as "digital read" (`l_dr`; see example below). These functions take in a pointer to the Lua state and read/write variables by popping/pushing from a software stack in that Lua state.
2. Registering that C function during startup-up using `lua_register`.

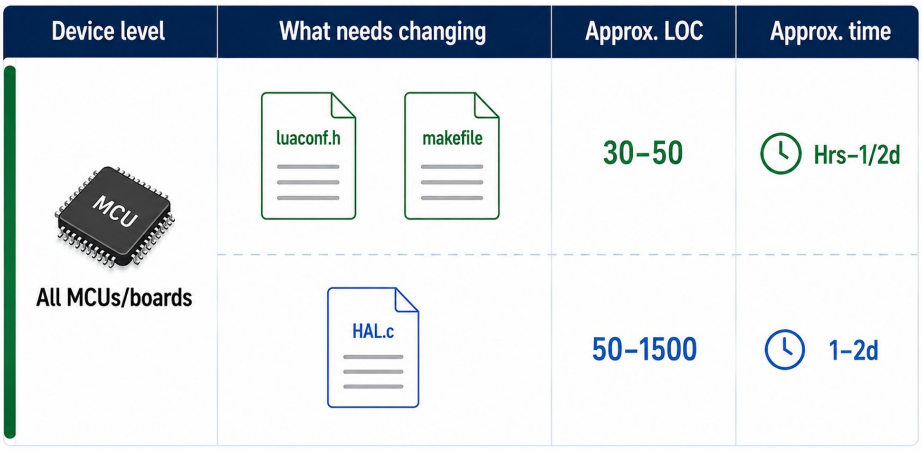


Now you can call the `dr` function from Lua (or enter it on the REPL) (e.g. `dr("C", 13)` to do a digital read of pin C13) and the Lua VM will call `l_dr` and return the value on that pin. A minimal HAL can be built using just a few dozen of these functions (think of the Arduino HAL). Notably, there is no specific API or contract that Lua requires these HALs to have (as is the case for MicroPython and Zephyr) so you can design them however you'd like. Additionally, the ease with which Lua integrates with C/C++ means its very easy to write things like interrupts, callbacks, or performance-critical code in C/C++ while the rest of the application remains in Lua.

This is how all external functions are added to a Lua instance: by registering them. A neat consequence of this is that you can create multiple `lua_State` variables, one per task in your system, and they become fully isolated entities. They cannot touch hardware or interact with any other tasks' memory space without having a registered function to do so. You can have one task/`lua_State` running your OTA updates and have it be completely separate from the task/`lua_State` running your application code.

———————————————————————————————————

Because of this the worst thing you can say about Lua, I think, is that every MCU requires a new port (though much of that code can be reused between MCUs). But put another way this is also Lua's advantage: porting is *consistent*, almost no matter the MCU, and it's consistently *less* effort than what would be needed to port MicroPython or Zephyr to a new MCU (the bottom three rows in the tables above). Getting to a REPL is fairly easy (see the steps above) and adding your own HAL need only take a few days more, depending on how many peripherals you want to use and how complex they are to configure and use.



Interfacing with C/C++ code is pretty straightforward; in fact, it's the exact process we followed above for exposing our HAL functions. You write a C function that follows the standard function signature (`int (*c_fcn_called_by_lua)(lua_State*L)`), pushing and popping your values into or from the software stack in that function, then register it with Lua (`lua_register(L, "fcn_name", c_fcn_called_by_lua);`).

Lua also uses dynamic memory allocation, but its possible to provide your own allocator and so have more control over how a heap/pool is used. Additionally, if certain programming features are entirely avoided in the application code (such as avoiding `string.format` or `table.insert`), then it's possible to restrict any heap usage to initialization only.

Forth

Forth is an old language (it predates *C*!) based entirely around stack operations: everything in Forth does some combination of

- Popping values off the stack,
- Pushing values onto the stack,
- Reading/writing memory (whose address is on the stack), or
- Manipulating the stack in a few ways (swapping the top two values in the stack, duplicating the top of the stack, etc)

The following Forth code prints the squares of the numbers 0 through 9 and defines a word (i.e. a function) to calculate the sum of two numbers squared.

```
: PRINT-SQUARES ( --
    10 0 DO
        I . ." squared is " I DUP * . CR
```

```
    LOOP
;

: SUM-OF-SQUARES ( a b -- c )
    SQUARE SWAP SQUARE +
;
```

This is... admittedly not the easiest way to code. It made sense 55 years ago when code was much closer to the hardware than it is today, but Forth really shows its age when you try to do anything involving more than three variables, such as operating on arrays, structs, or strings. The stack dance you'd need to do for a simple bubble sort would be mind-bending. By the same token, calling into C/C++ functions and exchanging data with them is challenging and essentially involves treating every data type as an array of Forth "words" or "cells" (typically a 32-bit number; though there are Forth operations that can treat two adjacent cells like the HIGH and LOW values of a combined 64-bit number). (Imagine being forced to think of every struct and string in C as an array of `uint8_t`.)

The main reason Forth is on this list is because it's so dang small! The final compiled size of zForth on my Nucleo-F411 was only **25 kB**, of which about 16 kB is attributable to zForth (the rest being the STM32 HAL, C library functions, and main). If you want an interpreted language that you can run on your 8-bit MCU with 32 kB of flash, this is it.

```
> ls -l build/zforth_repl_MCU.bin
-rwxrwxr-x 1 nathancharlesjones nathancharlesjones 26040 May 30 11:04 build/zforth_repl_MCU.bi
```

Based on my research, Forth runs about **5-20x slower** than C/C++.

Like, Lua there's not really any such concept as a "supported port" so you'll similarly need to provide your own HAL for each MCU you use. Like Lua, porting is therefore also highly predictable with fewer files to touch and leverages the same build system you use for the rest of your code.

zForth does not use the heap.

Conclusion

It turns out that each option is good at a few things, but not everything, and which one works best for a given project is very much dependent on the nature of that project and its developers. A side-by-side comparison of each of the options we've looked at reveals their relative strengths and weaknesses, though I want to reiterate that these values should be considered "rough heuristics" as opposed to "hard limits". In particular, each language's "relative speed" (relative to Zephyr, running native C/C++ code), is based only on some

preliminary research, not any empirical data, and it is **HIGHLY** dependent on the actual tasks being done, the data representation(s) being used, and how optimized the program/system is.

		MicroPython	**Zephyr**	**Lua**	**Forth**
Start-up time	"Fully-supported board"	Hrs-1/2d	Hrs-1/2d	REPL: Hrs-1/2d HAL: 1-2d	REPL: Hrs-1/2d HAL: 1-2d
	"New vendor"	1-8w	4-10w		
Interfacing with C/C++		Difficult (but still very doable)	Trivial (it *is* C!)	Easy (uses virtual stack), though one step removed from Zephyr	Easy for basic data types; weird for arrays, structs, strings, etc
Compiled size (kB)		277	44 (not including HAL)	144 (not including HAL)	25 (not including HAL)
Speed (relative)		50-250x	1x	10-50x	10-20x
Notes		• A+ language • Unparalleled library support • Requires heap • HAL is pre-defined (could be good or bad)	• Not interpreted • Strong support for various peripheral devices • HAL is pre-defined, but does expose access to each register	• A+ language • Secure microservices • Requires heap (but can be restricted to initialization with programmer discipline)	• Stack-ops are weird

In short, use **MicroPython** if:

- You want to use the full power of the MicroPython language and its extensive set of libraries
- You don't mind the speed or size and intend to write your entire application in Python
- You don't mind spending more time up-front on porting efforts

Use **Zephyr** if:

- You want the closest thing to pure C/C++ (small size, fast, easiest to interface with other C/C++ code)
- You don't care about not having an interpretable language
- You don't mind spending more time up-front on porting efforts (or you expect to amortize any porting efforts over the number of different MCUs you plan on targeting your application to in the future)

Use **Lua** if:

- You want to use a mature, interpretable language without the size of MicroPython
- You want something almost as fast as Forth
- You want to use this tool on many different MCUs and don't want porting efforts to take multiple weeks each time

Use **Forth** if:

- You need the smallest interpretable language possible
- You need something faster than Lua
- You don't mind using the stack/RPN (including for code that works on 3+ variables at a time)

Personally, I think Lua is the most compelling option on this list and I intend to explore it further on some upcoming projects. Regardless of my opinion, however, I hope you've found some use in this comparison and feel capable of making a more informed choice on your next project.

If you've made it this far, thanks for reading and happy hacking!